

成果证明报告

委托人: 方明俊

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检索结果:

经检索, 委托人方明俊提供的检索目录, 共有 1 篇论文被工程索引(EI)数据库收录。

(详细情况见附件)

特此证明



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EI 附件:

1. IVeri: A Scalable Privacy-Preserving Interdomain Configuration Verification Tool via Secure Multi-Party Computation

Accession number: 20253919245568

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Abstract:The fundamental challenge of configuration verification in an interdomain network is privacy because each autonomous system (AS) treats its network configuration files as private information and is not willing to share them with others. In this paper, we present IVeri, a scalable privacy-preserving interdomain configuration verification system based on secure multi-party computation. IVeri supports privacy-preserving verification and scalable verification via the following designs: (1) a verification algorithm that meets secure multi-party computation security requirements, (2) a data aggregator that reduces communication overhead while preserving privacy, (3) an algorithm that accelerates the simulation process based on routing algebra, and (4) an incremental verification algorithm that handles minor configuration changes. Extensive experiments with open-source datasets demonstrate that IVeri's optimization techniques significantly improve scalability, enabling the verification of large networks with 1000 ASes in under 3 hours, which outperforms state-of-the-art solutions. © 2025 IEEE.

Main heading:Network security

Controlled terms:['Differential privacy', 'Large datasets', 'Privacy by design', 'Privacy-preserving techniques']

Uncontrolled terms:['Configuration verification', 'Domain specific', 'Domain-specific security and privacy architecture', 'Inter-domain', 'Network reliability', 'Privacy preserving', 'Secure multi-party computation', 'Security and privacy', 'Verification algorithms', 'Verification tools']

Classification code:['1101.2 Machine Learning', '1106 Computer Software, Data Handling and Applications', '1106.2 Data Handling and Data Processing', '1108 Security and Privacy', '1108.1 Cybersecurity', '1108.2 Cryptography']

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